

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1243

Boeing Renton 2 100 kV
February 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1243. The Interconnection Customer proposes a 127.6 MW solar photovoltaic facility interconnecting at the Boeing Renton 2 100 kV substation in the Indiana area. The request was received on July 11, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 377 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$41,818,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hassen Cassel to Muldoon Mill 100.0 kV
Contingency: Wayne Trace to Darlington Road 100.0 kV line
Loading: 217.2% (Rating: 107 MVA)
DFAX: 5.4%
Pre-project loading: 210.9%
2. Facility: Darlington Road Substation to Hassen Cassel 100.0 kV
Contingency: Darlington Rd to Wayne Trace 100.0 kV
Loading: 186.5% (Rating: 118 MVA)
DFAX: 5.4%
Pre-project loading: 180.7%
3. Facility: Lake arion to Point A 100.0 kV line
Contingency: Dekalb to Richland 500.0 kV
Loading: 141.1% (Rating: 172 MVA)
DFAX: 5.7%
Pre-project loading: 136.9%
4. Facility: Roseau County to S Whitehall 100.0 kV
Contingency: John S Rainey to Trier 100.0 kV
Loading: 141.1% (Rating: 131 MVA)
DFAX: 11.7%
Pre-project loading: 129.7%
5. Facility: Barret to Point A 100.0 kV
Contingency: Richland to Dekalb 500.0 kV
Loading: 138.0% (Rating: 191 MVA)
DFAX: 5.7%
Pre-project loading: 134.2%

6. Facility: Wodburn to Monroeville 100.0 kV
Contingency: Richland to Dekalb 500.0 kV
Loading: 127.6% (Rating: 230 MVA)
DFAX: 5.6%
Pre-project loading: 124.5%
7. Facility: Sage to Wooburn 100.0 kV
Contingency: Richland to Dekalb 500.0 kV
Loading: 125.4% (Rating: 244 MVA)
DFAX: 5.6%
Pre-project loading: 122.5%
8. Facility: Barret to Roseau County 100.0 kV
Contingency: Richland to Dekalb 500.0 kV Line
Loading: 124.7% (Rating: 200 MVA)
DFAX: 10.4%
Pre-project loading: 118.0%
9. Facility: Sage Substation to Monte Del Estado 6303 100.0 kV line
Contingency: Dekalb to Richland 500.0 kV
Loading: 122.7% (Rating: 264 MVA)
DFAX: 5.6%
Pre-project loading: 120.0%
10. Facility: Segs 5 to Pevine 100.0 kV
Contingency: Richland to Dekalb 500.0 kV
Loading: 113.2% (Rating: 203 MVA)
DFAX: 14.5%
Pre-project loading: 104.1%
11. Facility: Wayne Trace to Allen 100.0 kV Line
Contingency: Segs 5 Substation to Hillcrest 100.0 kV
Loading: 105.4% (Rating: 390 MVA)
DFAX: 7.2%
Pre-project loading: 103.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hassen Cassel to Muldoon Mill 100.0 kV	Rebuild line with higher-capacity conductor	\$3,500,000	34% (\$1,193,000)
Hassen Cassel to Darlington Rd 100.0 kV	Rebuild line with higher-capacity conductor	\$1,621,000	34% (\$553,000)
Pt A to Lake Marion 100.0 kV	Rebuild line with higher-capacity conductor	\$3,915,000	36% (\$1,427,000)
S Whitehall to Roseau County 100.0 kV	Rebuild line with higher-capacity conductor	\$14,463,000	100% (\$14,463,000)
Barret to Pt A 100.0 kV line	Rebuild line with higher-capacity conductor	\$15,817,000	36% (\$5,764,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monroeville to Woodburn 100.0 kV	Rebuild line with higher-capacity conductor	\$12,298,000	32% (\$3,975,000)
Sage to Wooburn 100.0 kV	Rebuild line with higher-capacity conductor	\$3,753,000	32% (\$1,213,000)
Barret to Roseau County 100.0 kV line	Rebuild line with higher-capacity conductor	\$9,771,000	100% (\$9,771,000)
Sage Substation to Monte Del Estado 6303 100.0 kV line	Rebuild line with higher-capacity conductor	\$4,250,000	32% (\$1,374,000)
Segs 5 to Pevine 100.0 kV	Rebuild line with higher-capacity conductor	\$3,867,000	51% (\$1,981,000)
Wayne Trace to Allen 100.0 kV Line	Reconductor line	\$1,341,000	8% (\$104,000)
	Total allocated to project		\$41,818,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.